

Federated Drift-Aware Graph Neural Forecasting for Real-Time Passenger Flow

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Abstract

This paper introduces FedST-GNN, a federated spatio-temporal graph neural network that combines encrypted federated averaging with frequency-domain Transformers and ADWIN-triggered meta-learning for privacy-preserving passenger flow forecasting. Evaluation on the Copenhagen-Flow dataset (18.7M events, 312 stops) demonstrates 5% and 7% improvements in MAE and RMSE respectively, over Temporal Fusion Transformer baselines, while maintaining 38ms inference latency on commodity hardware. During anomalous events such as a city half-marathon, the adaptive drift detection mechanism reduced peak prediction errors by 41% within a 5MB communication budget per federated round. These results establish that federated graph neural architectures can deliver real-time accuracy for intelligent transportation systems while preserving data privacy.

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1. Introduction

Accurate passenger flow forecasting is critical for dynamic transit operations, yet urban mobility exhibits increasing volatility from extreme weather, incidents, and mass events that can spike forecast errors within minutes of regime shifts [1] [2]. While Spatio-Temporal Graph Neural Networks (ST-GNNs) like STGCN [3] and ASTGCN [4] capture complex spatial-temporal dependencies better than classical statistical models. Most deployments centralise raw smart-card data, violating data sovereignty regulations. Moreover, the intersection of federated learning and concept drift for passenger flow forecasting remains underexplored.

This paper introduces FedST-GNN, a drift-aware federated ST-GNN that combines graph attention with frequency-domain Transformers and ADWIN-triggered meta-learning for privacy-preserving real-time forecasting. We evaluate our approach on the new Copenhagen-Flow dataset comprising 18.7M passenger events across 312 stops (2022–2024) with weather and event metadata.

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2. Literature Review

Classical approaches like SARIMA achieve reasonable performance during stable periods but suffer during anomalous events, with MAPE exceeding 25% during demand spikes [5]. Deep learning methods have shown substantial improvements, with LSTM reducing RMSE by 18% over statistical baselines despite higher computational costs [6]. Graph neural networks further advance the field by jointly modelling spatial topology and temporal dynamics [7]. STGCN achieves 11% MAE reduction over LSTM on subway data, while DCRNN's bidirectional diffusion operators provide an additional 4 percentage point gains. Recent frequency-domain innovations like Informer reduce temporal attention complexity to $O(L \log L)$ while achieving 7 percentage point RMSE improvements on long sequences [8]. Most federated learning research in mobility focuses on ride-hailing or generic traffic prediction, with limited exploration of passenger flow forecasting [9]. FedProx demonstrates that proximal regularisation can mitigate client heterogeneity, but the interaction between federation and concept drift remains unquantified [10]. Privacy mechanisms typically rely on secure aggregation, though differential privacy approaches incur 3–4 percentage points MAE penalties, highlighting the privacy-utility tradeoff [11]. Signal-based detectors like ADWIN provide efficient drift detection through sliding window comparisons, while meta-learning approaches like Reptile enable rapid adaptation with few gradient steps, reducing post-drift [12] MAE by 26% on time series data [13]. However, systematic evaluation of drift adaptation within federated graph neural architectures for passenger flow data is absent from current literature.

3. Materials and Methods

The Copenhagen-Flow dataset aggregates automatic fare collection and vehicle location data from Movia, the regional transport authority, spanning January 2022 to December 2024. The dataset comprises 18.7M passenger events across 312 stops and 42 bus routes, plus 3 metro lines, with meteorological data from the Danish Meteorological Institute spatially joined via Voronoi tessellation. After removing erroneous GPS matches (0.47% through four-sigma headway filtering), the dataset provides minute-level observations with mean boardings of 17.4 passengers per minute ($\sigma = 31.2$) [15]. Each stop-time observation is encoded as a 10-dimensional feature vector including boardings, alightings, headway, temperature, precipitation, event flags, and cyclical encodings for time-of-day and day-of-week patterns. All continuous variables are z-normalised per training split.

3.1. Graph Construction and Problem Formulation

The dynamic graph connects stops via directed edges if they are consecutively served on routes or within 750 meters Euclidean distance. Edge weights combine geographic decay and time-varying congestion using an exponential distance decay function multiplied by the ratio of current headway to average network headway, where the distance decay parameter is set to 350 meters and average headway represents the long-run network baseline. Given a sliding window of 12 minutes, the forecasting task estimates passenger load one minute ahead using pinball loss for quantile regression at the 10th, 50th, and 90th percentiles.

3.2. FedST-GNN Architecture and Federated Training Protocol

The architecture integrates three key components: (1) spatial encoding via GATv2 attention mechanisms that compute node embeddings through weighted neighbour aggregation, (2) frequency-domain transformers that apply multi-head attention in Fourier space, reducing temporal complexity from $O(L^2)$ to $O(L \log L)$, and (3) quantile regression heads mapping embeddings to conditional quantiles via two-layer MLPs. The federated protocol uses FedAvg encrypted with Top-K gradient sparsification ($K = 0.05$) to limit communication to 5MB per 15-minute round. Each client performs local gradient computation with ADWIN-triggered meta-learning for drift adaptation, executing two rapid inner-loop updates when distributional shifts are detected. Global model updates aggregate sparse gradients weighted by local sample counts.

4. Results and Discussion

FedST-GNN achieves superior performance across all metrics, with MAE of 2.43 and RMSE of 3.12, representing 5.4% and 7.1% improvements over the strongest baseline (TFT), respectively. Diebold-Mariano tests confirm statistical significance ($p < 0.05$) for all comparisons, with test statistics ranging from -2.84 to -6.27. The model maintains approximately linear error growth across 30-minute prediction horizons ($\partial MAE / \partial h \approx 0.037$), contrasting with super-linear escalation in transformer baselines beyond 16 minutes. Probabilistic calibration analysis reveals well-calibrated uncertainty quantification, with an integrated Brier score of 3.2×10^{-3} —27% better than TFT. This superior calibration provides reliable confidence intervals essential for operational decision-making in transit systems. Table 2 reports aggregate error metrics. FedST-GNN attains the smallest error in every column, surpassing the strongest baseline (TFT) by 5.4% in MAE and 7.1% in RMSE. Bootstrapped 95% confidence intervals (1,000 resamples) are shown in parentheses.

Table 1. Aggregate forecasting accuracy (Oct–Dec 2024). Best results are **bold**; second best are underlined.

Model	MAE	RMSE	sMAPE (%)	CRPS
ARIMA	3.90 (0.05)	4.88 (0.07)	13.07	2.31
Prophet	3.75 (0.04)	4.63 (0.06)	12.52	2.24
XGBoost	2.95 (0.03)	3.68 (0.04)	10.01	1.72
LSTM	2.88 (0.03)	3.59 (0.04)	9.86	1.69
GRU	2.85 (0.03)	3.55 (0.04)	9.73	1.66
STGCN	2.69 (0.03)	3.39 (0.04)	9.12	1.58
DCRNN	<u>2.61</u> (0.03)	3.26 (0.04)	<u>8.94</u>	<u>1.54</u>
TFT	2.57 (0.03)	<u>3.24</u> (0.04)	8.98	1.55
FedST-GNN	2.43 (0.02)	3.12 (0.04)	8.46	1.47

4.1. Concept Drift Adaptation and Architectural Contributions

During a city-wide half-marathon disruption, the ADWIN detector identified distributional shifts within seven minutes—faster than traditional Page-Hinkley methods requiring 10-15 minutes. The two-step MAML adaptation achieved 41% error reduction ($t = -11.8, p < 10^{-10}$), demonstrating effective real-time adaptation to non-stationary passenger flow patterns. Ablation studies reveal synergistic effects of core components: removing frequency-domain attention increases MAE by 4.9%, eliminating graph connectivity raises error by 7.2%, and combined removal results in 11.8% degradation. The frequency-domain mechanism achieves efficient periodic modelling with 18% fewer parameters than comparable spectral transformers while preserving long-range temporal dependencies.

4.2. Computational Efficiency and Operational Viability

FedST-GNN achieves 1,600 requests/second throughput with 38ms median latency, well within the 125ms operational threshold. The memory consumption of the GPU of 3.9 GiB represents a reduction of 17% versus TFT, allowing edge deployment. The energy efficiency of 0.78 J / request provides an improvement of 24%, reducing operational costs and the carbon footprint.

Table 2. Resource footprint during peak load.

Metric	FedST-GNN	TFT	Relative (%)
Median latency (ms)	38	44	−13.6
P99 latency (ms)	82	104	−21.2
GPU util. (%)	71	78	−9.0
GPU memory (GiB)	3.9	4.7	−17.0

5. Conclusions

This study developed and evaluated FedST-GNN, a federated, drift-aware spatio-temporal graph neural network for real-time passenger flow forecasting. The architecture successfully integrates graph attention for spatial dependencies, frequency-domain transformers for efficient temporal modelling, encrypted federated averaging with Top-K sparsification for privacy preservation, and ADWIN-triggered meta-learning for concept drift adaptation. Key limitations include single-city evaluation scope, absence of formal differential privacy guarantees, homogeneous client assumptions, and limited external feature integration. Future work will address multi-city validation, Rényi differential privacy integration, heterogeneous federated learning approaches, and closed-loop optimisation with reinforcement learning agents.

5.1. Key Contributions and Outcomes

The experimental evaluation on the Copenhagen-Flow dataset demonstrates several significant achievements. FedST-GNN reduced mean absolute error by 5.4% and root-mean-square error by 7.1% compared to the strongest baseline while maintaining superior probabilistic calibration. The drift adaptation mechanism achieved 41% error reduction during disruption events, with recovery occurring within seven minutes of detection. The system maintains operational viability with 40ms median latency at 1,600 queries per second and communication overhead limited to 5MB per federation round. These results establish FedST-GNN as a viable solution for privacy-compliant transit analytics that meets real-world operational requirements. The combination of federated learning with drift adaptation addresses critical challenges in modern transportation systems where data sovereignty and dynamic adaptation are essential. The public release of data, code, and reproducibility scripts provides an open benchmark for advancing privacy-preserving transit analytics research.

5.2. Future Research Directions

The framework opens several promising research avenues, including multi-city generalisation studies, integration of formal differential privacy guarantees, heterogeneous federated learning approaches, and closed-loop optimisation with reinforcement learning. The established benchmark facilitates comparative evaluation of future federated graph learning methods in transportation domains. FedST-GNN advances the state-of-the-art in privacy-preserving, drift-resilient passenger flow forecasting while providing a reproducible foundation for intelligent transportation systems research. The demonstrated ability to maintain centralised-level accuracy under federated constraints with real-time adaptation capabilities establishes a new paradigm for privacy-compliant urban mobility analytics.

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